

Belief Networks

Video Lesson

Duration: 00:28:09

Slide Deck

M11S07 (<https://canvas.tamu.edu/courses/430127/files/92570693?wrap=1>) 

Lesson Transcript

In this final lecture of this module, we will look into belief networks.

Probabilistic Reasoning

Belief Network represent the dependence between random variables, and give a concise specification of the joint probability distribution. It is represented as a directed acyclic graph (DAG):

- 1 a set of random variables : nodes of the network
- 2 a set of directed edges from one node to another
- 3 each node has a conditional probability table that quantifies the effect the parents have on that node. The parents are the nodes pointing to that node.
- 4 the graph has no cycles

Belief networks enable more intuitive and more efficient probabilistic reasoning.

Belief networks represent the dependence between random variables and give a concise specification of the joint public distribution. It is represented as a directed acyclic graph or a DAG.

To begin, we have a set of random variables, and these form the nodes of the network or the graph. We have a set of directed edges from one node to another, indicating the dependence between these two different random variables. Each node has a conditional probability table that quantifies the effect of the parents on that particular node. The parents are the nodes pointing the arrow to that node. Again, to emphasize, the graph has no cycles.

Belief Network: Example



New burglar alarm was installed.

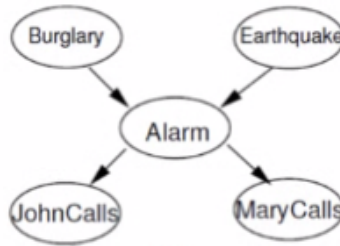
- The alarm can be triggered by either an actual burglary or an earthquake.
- Neighbors John and Mary agreed to call you at work when they hear the alarm.

Here is a simple example domain represented as a belief network. Suppose we installed the new burglar alarm. The alarm can be triggered by either an actual burglary or an earthquake, as shown

by these two arrows. And we set it up so that when our neighbors, John and Mary, hear the alarm from our home, they will give us a call at our work.

Here we see the random variables, *burglary*, *earthquake*, *alarm*, *JohnCalls*, and *MaryCalls*. Based on these arrows, we can see the dependencies.

Belief Network: Example (cont'd)

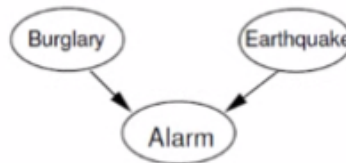


Example question: If you got calls from John and Mary, what is the chance of it being a totally false alarm (not a burglary, nor an earthquake)?
You can ask any conjunctive combination.

With this belief network, we can ask questions like this. If we got calls from both John and Mary, what is the chance of it being a totally false alarm? Meaning that there was an alarm, but there was no burglary or earthquake.

We can ask for any such conjunctive combination of events using the Belief Network.

Belief Network: Example (cont'd)



Each node has a conditional probability table fully describing $P(\text{Current} | \text{Parent}_1, \text{Parent}_2, \dots, \text{Parent}_n)$:

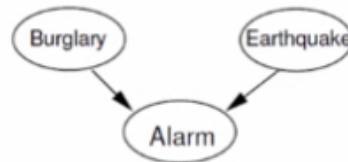
Burglary	Earthquake	P	$\neg P$
True	True	0.950	0.050
True	False	0.950	0.050
False	True	0.290	0.710
False	False	0.001	0.999

$P = P(\text{Alarm} | \text{Burglary}, \text{Earthquake})$

As mentioned earlier, each node or each random variable has its own conditional probability table

like this.

Belief Network: Example (cont'd)



Each node has a conditional probability table fully describing $P(\text{Current} | \text{Parent}_1, \text{Parent}_2, \dots, \text{Parent}_n)$:

Burglary	Earthquake	P	$\neg P$
True	True	0.950	0.050
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False	True	0.290	0.710
False	False	0.001	0.999

$P = P(\text{Alarm} | \text{Burglary}, \text{Earthquake})$

$P(A|B,E)$

$P(\neg A | \neg B, E)$

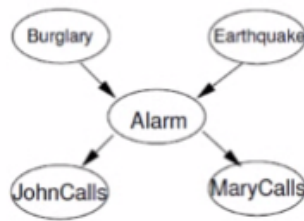
This conditional probability table down here is the conditional probability table for this alarm random variable.

These numbers indicate what is the probability of the alarm being true given these two different conditions. For example, this one corresponds to the probability of *alarm* given *burglary* and *earthquake*. Here we are using the same binary random variable notation. So, A means $A = \text{True}$, B means $B = \text{True}$, and so on.

Let us consider this one. This column corresponds to the alarm not sounding, meaning *alarm* = **False**. This particular value would correspond to the probability of $\neg A$, given $\neg B \wedge E$. Again, we use the same binary random variable notation.

These parent nodes, *burglary* and *earthquake*, also have their own conditional probability table, but because no arrows are pointing to them, it simply becomes the prior probability. When *earthquake* is **True**, the probability of an earthquake is 0.002. The probability of an earthquake being **False** is 0.998 as these have to sum up to 1.

Semantics of Belief Networks



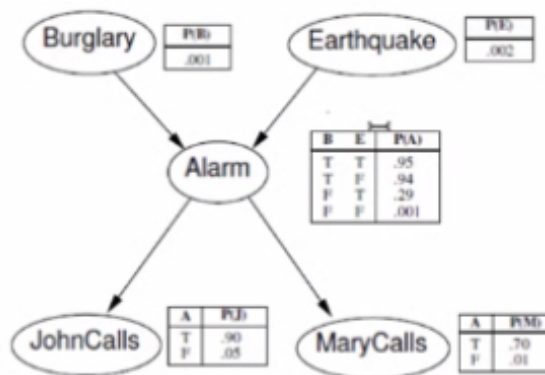
The network can be viewed as

- a representation of the **joint probability distribution** (this view is helpful when **constructing** the network), or
- an encoding of a collection of **conditional independence** statements (this view is helpful when designing effective **inference procedures**).

The semantics of belief networks can be viewed in two ways. One is a representation of the full joint probability distribution, and the next is the belief network encoding a collection of conditional independent statements.

The first view is helpful when constructing the network, and the second view is helpful when designing effective inference procedures.

Belief Network: Representing Joint Prob. Dist.

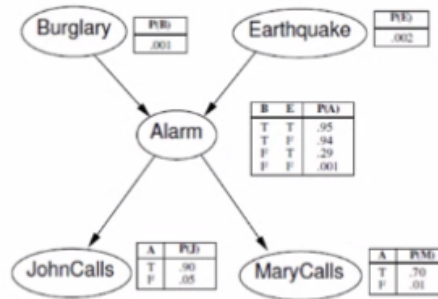


Each row in the conditional probability tables are:

$$\begin{aligned} P(X_1 = x_1, \dots, X_n = x_n) &= P(x_1, x_2, \dots, x_n) \\ &= \prod_{i=1}^n P(x_i | \text{Parent}(X_i)) \end{aligned}$$

Let us see how this belief network, with all of the conditional probability tables and the prior probabilities, can define the full joint probability distribution.

Belief Network: Representing Joint Prob. Dist.



Each row in the conditional probability tables are:

$$P(X_1 = x_1, \dots, X_n = x_n) = P(x_1, x_2, \dots, x_n)$$

$$= \prod_{i=1}^n P(x_i | \text{Parent}(X_i))$$

Handwritten table showing the joint probability distribution $P(B, E, A, J, M)$:

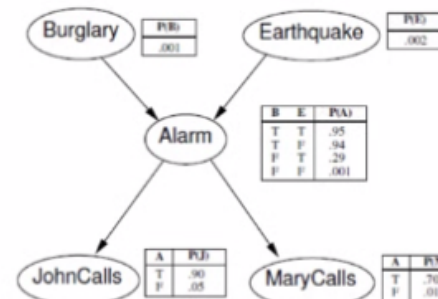
B	E	A	J	M	$P(B, E, A, J, M)$
T	T	T	T	T	
T	T	T	T	F	
T	T	F	T	T	
T	T	F	T	F	
T	T	F	F	T	
T	T	F	F	F	
T	F	T	T	T	
T	F	T	T	F	
T	F	T	F	T	
T	F	T	F	F	
T	F	F	T	T	
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F	F	T	F	T	
F	F	T	F	F	
F	F	F	T	T	
F	F	F	T	F	
F	F	F	F	T	
F	F	F	F	F	

Normally with five random variables, to describe the full joint probability distribution, we need 2^5 rows, that is, 32 rows.

However, in this belief network, we need 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, that is, only 10 rows in the totality of all the conditional probability and prior probability tables.

Although it is much more compact, we can still use this information to reconstruct the full joint probability distribution using this formula down here.

Belief Network: Representing Joint Prob. Dist.



Each row in the conditional probability tables are:

$$P(X_1 = x_1, \dots, X_n = x_n) = P(x_1, x_2, \dots, x_n)$$

$$= \prod_{i=1}^n P(x_i | \text{Parent}(X_i))$$

Handwritten formula showing the joint probability distribution as a product of conditional probabilities:

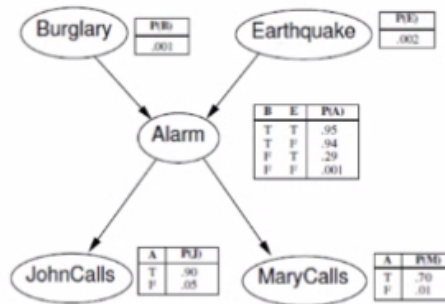
$$P(B | \text{Parents}(B)) \times P(E | \text{Parents}(E)) \times P(A | \text{Parents}(A)) \times P(J | \text{Parents}(J)) \times P(M | \text{Parents}(M))$$

Because this is the product for this particular example, this will become the probability of B given parents of B , the probability of E given parents of E .

Probability of A given parents of A , times probability of J given parents of J , times probability of

M given parents of M .

Belief Network: Representing Joint Prob. Dist.



Each row in the conditional probability tables are:

$$\begin{aligned}
 P(X_1 = x_1, \dots, X_n = x_n) &= P(x_1, x_2, \dots, x_n) \\
 &= \prod_{i=1}^n P(x_i | \text{Parent}(X_i))
 \end{aligned}$$

Handwritten notes showing the joint probability distribution for the belief network:

$$\begin{aligned}
 &P(B) \\
 &P(E) \\
 &P(A | B, E) \\
 &P(J | A) \\
 &P(M | A)
 \end{aligned}$$

Now, explicitly considering these, this becomes $P(B)$ because B has no parents. This becomes $P(E)$ because E does not have any parents. So, this can be removed. As for $alarm$, its parents are $burglary$ and $earthquake$. So, these parents of A become B and E . For $JohnCalls$, the parent is only $alarm$.

This becomes the probability of John given A , $alarm$. The last one, the probability of Mary given its parents, becomes the probability of Mary given $alarm$.

Belief Network: Representing Joint Prob. Dist. (cont'd)

Handwritten notes showing the joint probability distribution for the belief network:

$$\begin{aligned}
 P(X_1 = x_1, \dots, X_n = x_n) &= P(x_1, x_2, \dots, x_n) \\
 &= \prod_{i=1}^n P(x_i | \text{Parent}(X_i))
 \end{aligned}$$

Handwritten assignment: T, F, T, T, F for Burglary, Earthquake, Alarm, JohnCalls, MaryCalls respectively.

Handwritten calculation: $P(A | B, \neg E)$

$\text{Parent}(X_j)$ refers to the event when $X_j = x_j$.

Imagine a case where each node in the example has a **True** or **False** assignment. X_j will then be either **True** or **False** for all j .

The belief network fully defines a joint probability distribution!

Here we have to be careful because depending on the particular combination of **True** and **False**

values for these random variables, we have to compute or look up the appropriate conditional probability when we are doing the multiplication. Let us consider the random variable *alarm*. So, this probability of X_i given parent X_i would become the probability of *alarm* because *alarm* is **True**. And given burglary $B \wedge \neg E$, because burglary is **True** and *earthquake* is **False** in this joint probability.

Calculating Probability of a Joint Event



Calculate the probability of the event that the alarm (A) has sounded but neither a burglary ($\neg B$) nor an earthquake ($\neg E$) occurred, and both John (J) and Mary (M) call:

$$\begin{aligned}
 &P(J \wedge M \wedge A \wedge \neg B \wedge \neg E) \\
 &= P(J|Prnts(J))P(M|Prnts(M))P(A|Prnts(A))P(\neg B)P(\neg E) \\
 &= P(J|A)P(M|A)P(A|\neg B \wedge \neg E)P(\neg B)P(\neg E)
 \end{aligned}$$

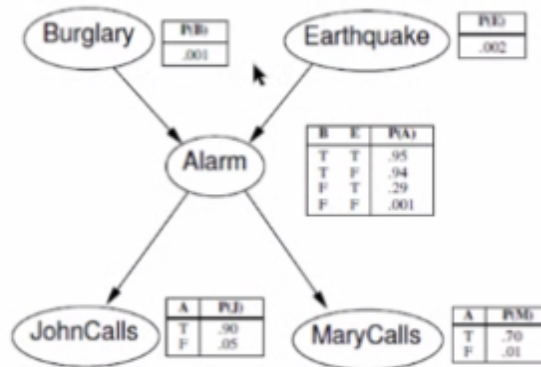
Let us look at this example. We want to calculate the probability of the event that the alarm sounded, that is, A , but neither a burglary, that is, $\neg B$, nor an earthquake occurred, that is, $\neg E$. And both John and Mary call. That is J and M .

When we write the joint probability, this becomes the probability of J and M and A and $\neg B$ and $\neg E$. So, we find the product of J , given parents of J , M given parents of M , A given parents of A , and $\neg B$ given the parents of B , but there is no parent, so it just becomes P of $\neg B$. The same with *earthquake*, it becomes the probability of $\neg E$. And we have to make sure that the particular sign here is always the same, which is what is given.

The parents of J become *alarm*. And because *alarm* was **True**, the probability of J given parents of J becomes the probability of J given A . Likewise, the probability of M given parents of M becomes the probability of M given A , A being the only parent.

Here is the interesting part. The probability of A given parents of A becomes the probability of A given $\neg B$ and $\neg E$. Because B and E were **False**, we have to put in the negations here appropriately. And finally, we have the same prior probabilities.

Calculating Probability of a Joint Event (cont'd)

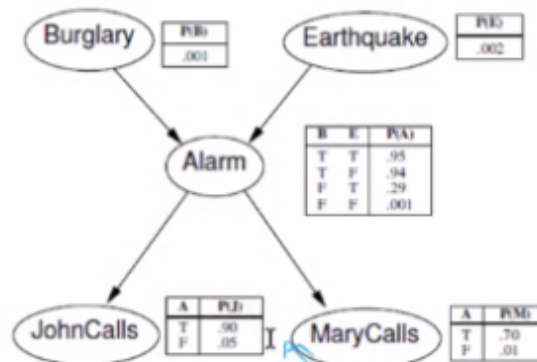


$$P(J|A)P(M|A)P(A|\neg B \wedge \neg E)P(\neg B)P(\neg E)$$

$$= 0.9 \times 0.7 \times 0.001 \times 0.999 \times 0.998 = 0.00062$$

Continuing from the previous slide, we can look up all of these values from the conditional probability tables and the prior probabilities. The probability of J given A would be this. And $A = \text{True}$. So, that will be 0.9.

Calculating Probability of a Joint Event (cont'd)



$$P(J|A)P(M|A)P(A|\neg B \wedge \neg E)P(\neg B)P(\neg E)$$

$$= 0.9 \times 0.7 \times 0.001 \times 0.999 \times 0.998 = 0.00062$$

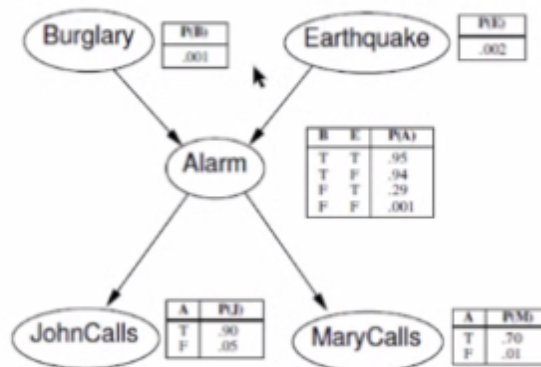
$$\begin{array}{c|c}
 A & P(J) \\
 \hline
 T & 0.10 \\
 F & 0.95
 \end{array}$$

Note that these conditional probability tables or the prior probability tables just show the minimal information from which we can get the full information. For example, here, the probability of not

JohnCalls given *A* is not shown, but we can quickly compute that from the information shown here. The probability of not *JohnCalls* given *alarm* being **True** is $1 - 0.9$. That would be 0.1.

The probability of not *JohnCalls* given no *alarm* would be $1 - 0.05$, which will give us 0.95.

Calculating Probability of a Joint Event (cont'd)



$$P(J|A)P(M|A)P(A|\neg B \wedge \neg E)P(\neg B)P(\neg E)$$

$$= 0.9 \times 0.7 \times 0.001 \times 0.999 \times 0.998 = 0.00062$$

Let us return to the computation P of J given A is 0.9. $P(M)$ given A is 0.7.

$P(A)$ given $\neg B$, and $\neg E$. We must look up the particular row where B is **False** and E is **False**. Then that will give us 0.001, which is that number here.

Then, when we look up $P(\neg B)$, it is not shown here. But we can compute that by subtracting this from 1, where we get 0.999. Then this, here, the probability of $\neg E$, is again $1 - 0.002$. So, we get 0.998.

As a result, when we multiply all of these, we get 0.00062.

Joint Probability Distribution Under Conditional Independence

In Belief Networks, the joint probability is given as follows:

$$P(x_1, \dots, x_n) = \prod_{i=1}^N P(x_i | \text{Parents}(X_i)).$$

This is derived from the two following equations:

$$\begin{aligned} P(x_1, \dots, x_n) &= P(x_n | x_{n-1}, \dots, x_1) P(x_{n-1}, \dots, x_1) \\ &= P(x_n | x_{n-1}, \dots, x_1) P(x_{n-1} | x_{n-2}, \dots, x_1) \\ &\quad \cdot \dots \cdot P(x_2 | x_1) P(x_1) \\ &= \prod_{i=1}^N P(x_i | x_{i-1}, \dots, x_1) \quad (2) \\ P(X_i | X_{i-1}, \dots, X_1) &= P(X_i | \text{Parents}(X_i)) \quad (3) \end{aligned}$$

Earlier, we saw that the full joint probability can be computed by the product of the different conditional probabilities and prior probabilities in the belief network. So, what allows us to do this kind of simplification?

Joint Probability Distribution Under Conditional Independence

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$$\begin{aligned} P(A, B, C, D) &= P(A | B, C, D) P(B, C, D) \\ P(A, B) &= P(A | B) P(B) \end{aligned}$$

We simply use this result based on the definition of conditional probability. So, P of A, B, C, D equals P of A given B, C, D times P of B, C, D .

In a simpler form, we have this. Here, we just have more variables on the right-hand side of this bar.

Joint Probability Distribution Under Conditional Independence

In Belief Networks, the joint probability is given as follows:

$$P(x_1, \dots, x_n) = \prod_{i=1}^N P(x_i | \text{Parents}(X_i)).$$

This is derived from the two following equations:

$$\begin{aligned}
 P(x_1, \dots, x_n) &= P(x_n, x_{n-1}, x_{n-2}, \dots, x_2, x_1) \\
 &= P(x_n | x_{n-1}, \dots, x_1) P(x_{n-1}, \dots, x_1) \\
 &= P(x_n | x_{n-1}, \dots, x_1) P(x_{n-1} | x_{n-2}, \dots, x_1) \\
 &\quad \dots P(x_2 | x_1) P(x_1) \\
 &= \prod_{i=1}^N P(x_i | x_{i-1}, \dots, x_1) \tag{2} \\
 P(x_i | x_{i-1}, \dots, x_1) &= P(x_i | \text{Parents}(X_i)) \tag{3}
 \end{aligned}$$

Let us start with this probability of $x_1, x_2, \dots, x_n$.

As we have seen earlier, the ordering does not matter, so let us rearrange that in reverse order. This is the same as probably of x_n, x_{n-1}, x_{n-2} , and so on. Then we use the same trick that we just saw to get the conditional probability of x_n given the rest and the probability of the rest multiplied. So, this is, by definition, the same.

Then we apply that same trick to the probability of x_{n-1} up to x_1 . So, that will become the probability of x_{n-1} given x_{n-2} , the rest, multiplied by P of x_{n-2} and the rest. If we keep on doing it, then we will end up here at the last part.

This will become x_2 given x_1 multiplied by P of x_1 .

Joint Probability Distribution Under Conditional Independence

In Belief Networks, the joint probability is given as follows:

$$P(x_1, \dots, x_n) = \prod_{i=1}^N P(x_i | \text{Parents}(X_i)).$$

This is derived from the two following equations:

$$\begin{aligned}
 P(x_1, \dots, x_n) &= P(x_n | x_{n-1}, \dots, x_1) P(x_{n-1}, \dots, x_1) \\
 &= P(x_n | x_{n-1}, \dots, x_1) P(x_{n-1} | x_{n-2}, \dots, x_1) \\
 &\quad \dots P(x_2 | x_1) P(x_1) \\
 &= \prod_{i=1}^N P(x_i | x_{i-1}, \dots, x_1) \\
 \hline
 P(x_i | x_{i-1}, \dots, x_1) &= P(x_i | \text{Parents}(X_i)) \quad \text{I}
 \end{aligned}$$

Exact computation (green arrow pointing to the first equation)

Simplifying step (pink wavy line under the second equation)

Handwritten notes on the right:

- $P(x_n | x_{n-1} \dots \dots x_1)$
- $P(x_{n-1} | x_{n-2} \dots \dots x_1)$
- $P(x_{n-2} | x_{n-3} \dots \dots x_1)$
- $P(x_2 | x_1)$
- $P(x_1)$

Equation (2) is the first line of the product, and Equation (3) is the last line.

If we write this out vertically, then it shows this kind of pattern. First, we have x_n given x_{n-1} up to x_1 . The next one is the probability of x_{n-1} given x_2 up to x_1 . So, this length will decrease by 1. It keeps on decreasing. When we reach the last line, we have just P of x_1 .

After this point, this product, or this product, is based on an exact computation. These are actually equivalent based on the axioms of probability.

This last part, which requires some assumption, will simplify the whole computation and give us this formula here.

This is all based on conditional independence. X_i is conditionally independent of everything else other than the parents given the parents.

Belief Network Construction

Given the nodes, we have to find which nodes are **directly** influenced by certain nodes, and from this, find out **conditional independence**.

- For example:

$$P(\text{MaryCalls} | \text{JohnCalls}, \text{Alarm}, \text{Earthquake}, \text{Burglary}) = P(\text{MaryCalls} | \text{Alarm}) \quad \text{I}$$
- So, even if *Earthquake* and *Burglary* can be found up-stream (e.g. *Earthquake* causing *Alarm* to go off, in turn causing *MaryCalls*), those events are conditionally independent from *MaryCalls*.

To construct a Belief Network, we need to find such dependency structure.

Now let us consider how we can construct these belief networks once we have the domain, once we have the random variables, and we have some kind of idea about how they depend on each other.

Given these nodes that correspond to the random variables, we have to find which nodes are directly influenced by certain nodes. From this, find out the conditional independence structure. For example, the probability of *MaryCalls* given *JohnCalls*, *alarm*, *earthquake*, and *burglary*, can be equated with the probability of *MaryCalls* given *alarm*.

That means that *MaryCalls* is conditionally independent of *JohnCalls*, *earthquake*, and *burglary* given *alarm*. Here the interesting thing is that even though *earthquake* and *burglary* can be found upstream of the alarm, these events are conditionally independent of *MaryCalls*. The reason is we asked Mary to call us when she hears the alarm, so she does not care about earthquakes or burglaries. The direct trigger for Mary calling us is the alarm sound.

To construct the belief network, we need to find such a dependency structure in our domain.

Belief Network Construction (cont'd)

The general procedure for Belief Network construction is as follows:

- ① Choose the set of relevant variables X_i that describe the domain.
- ② Choose an ordering of the variables.
- ③ While there are variables left:
 - ① Pick a variable X_i and add a node to the network for that variable.
 - ② Set $Parents(X_i)$ to some minimal set of nodes already in the net such that the conditional independence property is satisfied.
 - ③ Define the conditional probability table for node X_i .

Here is the general procedure for belief network construction. First, choose the set of variables, X_i , that describe the domain. Again, these should be the random variables. And choose an ordering of the variables. Which one will we add to the graph first?

And we go through this loop. While there are variables left in this ordered list, pick a variable X_i and add a node to the network.

Pick the parents of this X_i in the existing graph, and make them point to this particular variable that we just added.

These parents are some minimal setup nodes within the net, so the conditional independence

property is satisfied. Finally, we want to construct the conditional probability table for this random variable that we just added.

Evaluation of the Construction Algorithm

- Because newly added nodes cannot point to existing nodes, the resulting graph is always **acyclic**.
- Violation of axioms of probability is avoided.
- Compact, compared to the full joint probability table (**locally structured, or sparse**).
 - Belief network with n nodes (binary variables) and k parents per node has $n2^k$ entries in the conditional probability tables:

$$n \text{ nodes} \times 2^k \text{ per each node.}$$

- Full joint probability table: 2^n

Here are the properties of this construction algorithm.

Because newly added nodes cannot point back to the existing nodes, the resulting graph is always acyclic. As a consequence, a violation of axiom probabilities is avoided. And it gives us a compact probability table compared to the full joint probability table.

In the previous example of the *earthquake*, *burglary*, and *alarm* domain, we saw that we needed 32 entries in the full joint probability table, whereas we only needed 10 rows total in the conditional probability tables of the belief network.

In general, if the belief network has n nodes, that is, n binary random variables, and for each node, there is an average of k parents per node, then it has n times 2^k many rows in the conditional probability tables.

That is because each conditional probability table with k variables to account for has 2^k rows. And we have n many of those. So, it becomes n times 2^k .

As for the full joint probability table with n nodes, we need 2^n rows.

Evaluation of the Construction Algorithm

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 - Belief network with n nodes (binary variables) and k parents per node has $n2^k$ entries in the conditional probability tables:

n nodes $\times 2^k$ per each node.

$$n=100 \quad k=5$$

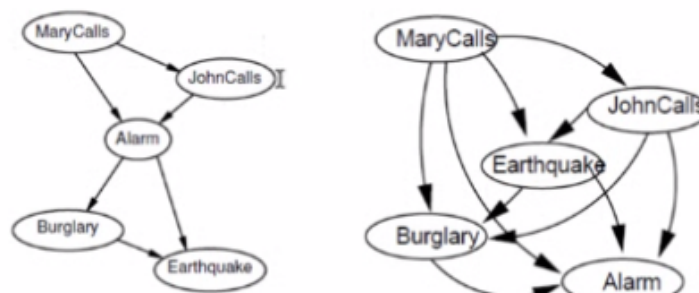
- Full joint probability table: 2^n

$$100 \times 2^5 = 3200$$

$$2^{100} \approx 1 \text{ mil}$$

To understand the huge difference, let us consider an n of 100 and a k of 5. For the belief network, that becomes 100 times 2^5 , which is 3,200, whereas the full joint probability table would be 2 to 100, which is about 1 million.

Importance of Node Ordering in Belief Network Construction



The resulting Belief Network can be vastly different when the order of insertion of nodes into the network is different. • *MaryCalls, JohnCalls, Alarm, Burglary, Earthquake*

- *MaryCalls, JohnCalls, Earthquake, Burglary, Alarm*

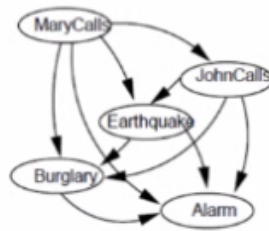
In the Belief Network construction algorithm, we saw that we need to order the nodes in a specific way. It turns out that this node ordering is very important.

Because we are adding one node at a time and connecting them, the resulting belief network can be vastly different. The order of insertion of the nodes into the network is different. Let us consider these two cases, *MaryCalls, JohnCalls, alarm, burglary, and earthquake*. That is one order. And the second ordering is *MaryCalls, JohnCalls, earthquake, burglary, and alarm*. If we follow the

construction algorithm for the first ordering, we get a belief network that looks like this, which is very strange. *MaryCalls* now has an arrow to *JohnCalls*.

For the second ordering, it gets even stranger. We have *MaryCalls* influencing everything, *burglary*, *earthquake*, *JohnCalls*, *alarm*, and *alarm* taking arrows from everything else, and so on. So, it is very complex.

Node Ordering and Joint Probability Tables



- Even with different graphs resulting from different node ordering, you can represent the same joint probability distribution!
- However, some represent the conditional independence relation much better than others.
- For example, the graph above requires the same number of entries as a full joint probability table.

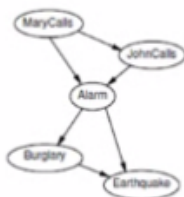
The interesting thing is that even though we have different ordering and different graphs resulting, these would essentially represent the same joint probability distribution.



But as we have seen, some of these versions, based on different ordering, may represent the conditional independence relation much better than others.

Also, this kind of suboptimal graph may require a lot of rows in the conditional probability tables, almost equaling the full joint probability table case.

Improper Node Ordering Can Cause Problems



- When adding nodes in the order of *MaryCalls*, *JohnCalls*, ..., conditional independence does not hold:

$$P(\text{JohnCalls} | \text{MaryCalls}) \neq P(\text{JohnCalls})$$

- Thus, *MaryCalls* has to become a parent of *JohnCalls*.
- This is because if Mary calls, it probably means that the alarm has gone off, so it makes it more likely that John calls.

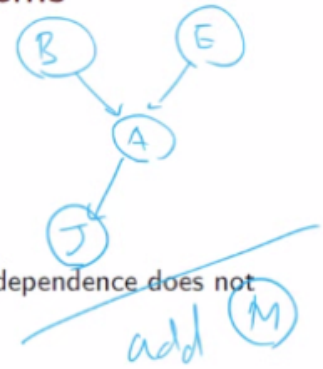
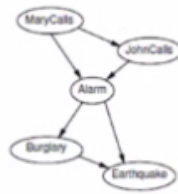
Why is it that when we add *MaryCalls*, and then it is time to add *JohnCalls*, we have to add this arrow?

This inequality summarizes the reason why we need this arrow. Let us say, initially, we are wondering about the probability of John calling us. Maybe it might be a low number, maybe 0.01 or something. But once Mary calls us, then our belief about John calling will actually go up because we arranged for them both to give us a call when there is an alarm sound.

Because Mary called us, our suspicion that John will also call might actually go up.

Because these two probability keys are not the same, we have actually to draw that arrow. But why was it the case that we did not need any arrow between *MaryCalls* and *JohnCalls* in the original graph?

Improper Node Ordering Can Cause Problems



- When adding nodes in the order of *MaryCalls*, *JohnCalls*, ..., conditional independence does not hold:

$$P(\text{JohnCalls} | \text{MaryCalls}) \neq P(\text{JohnCalls})$$

- Thus, *MaryCalls* has to become a parent of *JohnCalls*.
- This is because if Mary calls, it probably means that the alarm has gone off, so it makes it more likely that John calls.

$$p(M | J, A, B, E) = p(M | A)$$

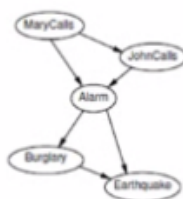
Let us consider this situation. In that graph, perhaps the ordering was *burglary*, *earthquake*, *alarm*, and *JohnCalls*, followed by *MaryCalls*. So, by the time we have these four nodes added, and we want to add *M*, then we want to find out where the probability of *MaryCalls* is given all the rest.

We can write this conditional probability of *MaryCalls* given *JohnCalls*, *alarm*, *burglary*, and *earthquake*.

But first, we note that *alarm* is the common cause of *JohnCalls* and *MaryCalls*, so at least we can exclude *JohnCalls*. So, we have the probability of *MaryCalls* given *alarm*. And as for this, these are not the direct cause of Mary calling us. So, we can also remove that. This is because we asked Mary to call us when she hears the alarm.

We did not instruct her to call us when she observed a burglary or an earthquake.

Improper Node Ordering Can Cause Problems



- When adding nodes in the order of *MaryCalls*, *JohnCalls*, ..., conditional independence does not hold:

$$P(\text{JohnCalls} | \text{MaryCalls}) \neq P(\text{JohnCalls})$$

- Thus, *MaryCalls* has to become a parent of *JohnCalls*.
- This is because if Mary calls, it probably means that the alarm has gone off, so it makes it more likely that John calls.

Because of this reason, *MaryCalls* must be a parent of *JohnCalls*.

To repeat, this is because if Mary calls, it is probably the case that the alarm has gone off, making it more likely that John will actually call.

Strategy for Better Node Ordering

- For a node *A* to be a parent of another node *B*, *A* must be added to the network before *B* is added.
- Thus, a node that has a direct influence on other nodes should be added to the network first, i.e. add the **root causes** first.

What is a good strategy for better node ordering? First, we observed that for node *A* to be a parent of another node *B*, *A* must be in the existing graph before *B* is added.

As a consequence, a node that has a direct influence on other nodes should be added to the network first. That is, the root causes go in the node list first.

Strategy for Better Node Ordering (cont'd)

When building the network, stick to a **causal model**, rather than the other way around (e.g. inferring cause given the effect). I

Causal model is beneficial because of these reasons:

- conditional probability tables can be made smaller
- the conditional probabilities can be easier to come up with
- easier to reason about the domain using the network

When building the network, we want to stick to a causal model rather than vice versa. Why is this beneficial? This causal model is beneficial because the conditional probability tables can be much smaller, and it would be much easier to estimate the conditional probabilities that go in these tables.

Finally, it is easier to reason about the domain using the network.

Probabilistic Inference

- **Diagnostic inferences:** $P(\text{Cause}|\text{Effect})$
- **Causal inferences:** $P(\text{Effect}|\text{Cause})$
- **Intercausal inferences:** causes of a common effect (**explaining away:** cause has already been found)
I
$$P(\text{Cause}|\text{Effect}) \gg P(\text{Cause}|\text{Effect} \wedge \text{OtherCause})$$
- **Mixed inferences:** combining two or more of the above

$$P(A|\text{CauseOfA} \wedge \text{EffectOfA})$$

In an earlier lecture in this module, we looked at diagnostic inference and causal inference. The diagnostic inference is the probability of a cause given an effect, and causal inference is the probability of an effect given a cause. There are other sources of inference: inter-causal inference and mixed inference. All of these could be done using the belief network.

Probabilistic Inference

- **Diagnostic inferences:** $P(\text{Cause}|\text{Effect})$
- **Causal inferences:** $P(\text{Effect}|\text{Cause})$
- **Intercausal inferences:** causes of a common effect (**explaining away:** cause has already been found)

$$P(B|A) \gg P(B|A \wedge E)$$

$$P(\text{Cause}|\text{Effect}) \gg P(\text{Cause}|\text{Effect} \wedge \text{OtherCause})$$

- **Mixed inferences:** combining two or more of the above

$$P(A|\text{CauseOfA} \wedge \text{EffectOfA})$$

In inter-causal inference, we have two causes, one cause and another cause of the same effect. So, initially, when we observe the effect, we have a belief about the initial cause. But after we get evidence, another cause could raise this effect. Then the probability or our belief about this original cause diminishes.

Let us look at this case. We heard the alarm, and we were wondering about the probability that there was a burglary. Probably this is very high, but later on, when we hear in the news that there was an earthquake, then this earthquake partially explains this alarm event. So, our belief about burglary being the cause of this alarm would actually decrease.

This conditional probability is much less than this one without this extra information about an earthquake. This is called inter-causal inference.

Probabilistic Inference

- **Diagnostic inferences:** $P(\text{Cause}|\text{Effect})$
- **Causal inferences:** $P(\text{Effect}|\text{Cause})$
- **Intercausal inferences:** causes of a common effect (**explaining away:** cause has already been found)

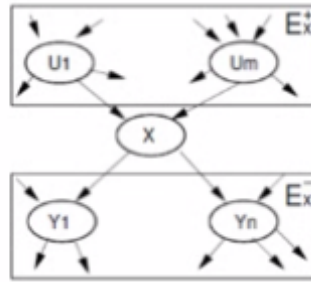
$$P(\text{Cause}|\text{Effect}) \gg P(\text{Cause}|\text{Effect} \wedge \text{OtherCause})$$

- **Mixed inferences:** combining two or more of the above

$$P(A|\text{CauseOfA} \wedge \text{EffectOfA})$$

Finally, we can mix causes and effects and have something called mixed inferences. What is the probability of A given cause of A and effect of A?

Answering Queries: A Brief Outline



Given a set of evidence E , find the conditional probability $P(X|E)$ where X is the query.

- Recursively determine the **causal support** E_x^+ for X .
- Recursively determine the **evidential support** E_x^- for X .

Note: This is only when the graph is **singly connected**.

Given a belief network, we can ask queries. For example, what is the probability of X given a set of pieces of evidence?

Here is the random variable X , and we have some parents and some children. So, the parents form the causal support, and the children form the evidential support.

To compute this conditional probability, we should recursively determine the causal support and recursively determine the evidential support. There are certain conditions for this to be valid. That is when the graph is singly connected. Many different inference procedures are available for the belief network, but that is beyond the scope of this course.

You can take a look at the textbook if you are interested.

Using Belief Networks and Probabilistic Inference

- Making decisions based on the derived probabilities and an agent's utility function.
- Deciding which additional variables to include in the model.
- Performing **sensitivity analysis** to find out which node is most important and thus should be more accurate.
- Explaining the results of reasoning.

Using the Belief Network, we can perform different forms of probabilistic inference. For example, we can make decisions based on the derived probabilities and an agent's utility function. This goes back to the beginning of this module, where we considered decision theory, and we can also decide which additional variables are needed in the model.

We could also perform sensitivity analysis. Which of these variables affect or which of these variables are more sensitive to different fluctuations? Once we identify these more sensitive nodes, then we can attempt to estimate the conditional probability table associated with that node to be more accurate.

Also, we can use the belief network to explain the reasoning results.

That is the end of this module. In the following module, we will look into machine learning.